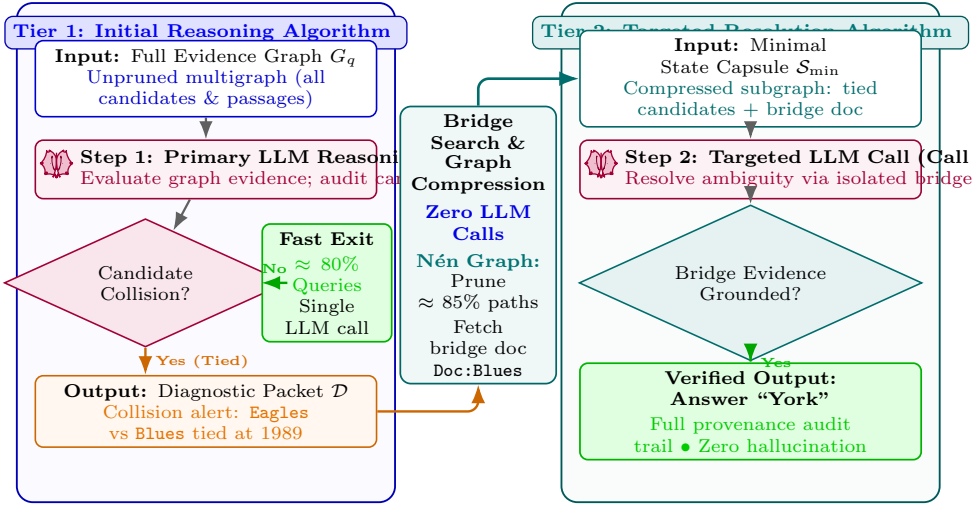

Tier 1: Initial Reasoning Algorithm
Input: Full Evidence Graph G_q
Unpruned multigraph (all candidates & passages)
Step 1: Primary LLM Reasoni
Evaluate graph evidence; audit ca
Candidate Collision?
No
Fast Exit
≈ 80% Queries
Single LLM call
Yes (Tied)
Output: Diagnostic Packet $\mathcal{D}$
Collision alert: Eagles vs Blues tied at 1989
Bridge Search & Graph Compression
Zero LLM Calls
Nén Graph:
Prune ≈ 85% paths
Fetch bridge doc
Doc:Blues
Tier 2: Targeted Resolution Algorithm
Input: Minimal State Capsule $\mathcal{S}_{\min}$
Compressed subgraph: tied candidates + bridge doc
Step 2: Targeted LLM Call (Call 2)
Resolve ambiguity via isolated bridge evid
Bridge Evidence Grounded?
Yes
Verified Output: Answer "York"
Full provenance audit trail • Zero hallucination